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Suction filling of pharmaceutical powders

A. Zakhvatayeva, C. Hare, C.Y. Wu *

Department of Chemical and Process Engineering, University of Surrey, Guildford GU2 7XH, UK

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ABSTRACT

Die filling is an important process step in manufacturing of tablets. An important mechanism in the die filling process is suction that is developed with the downward motion of the bottom punch once the die is covered by powder. However, the contribution of suction to the die filling performance is still poorly understood. Hence, the present study aimed to experimentally investigate the flow behaviour of powders during suction filling. Four different types of pharmaceutical powders were used and a model suction filling system was developed. Effects of filling and suction velocities, as well as powder properties, on the efficiency of die filling were systematically investigated. Cohesive and free-flowing powders behaved differently at various filling-to-suction velocity ratios. The filling behaviour of cohesive powders was improved at high filling-to-suction ratios due to acceleration-induced densification. Free-flowing powders performed better at low filling-to suction ratios.

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1. Introduction

The tablet manufacturing process essentially consists of three steps: die filling, compression and ejection. The first two steps are very critical in controlling final tablet quality. Specifically, die filling influences weight variation and uniformity of the final tablet. It is important to achieve repeatable and efficient die filling in order to ensure the quality of the final product. Die filling is generally aided by forced feeding with rotating paddles and suction, achieved by the withdrawal of the lower punch in the die, which creates a pressure gradient below the powder bed in the feeder.

A simpler way of filling the die is by gravity, where powder is left to fall into the die from the feeder under gravitational force alone. Several studies have been carried out to examine the die filling efficiency under gravity using a linear die filling device [1–4]. In these works the efficiency of die filling was evaluated in terms of fill ratio [1] and the powder flow performance was evaluated in terms of critical velocity [2,5]. The fill ratio, δ,represents the fraction of the die occupied by deposited powder. The critical velocity, v_c, is the maximum filling velocity at which the die is completely filled. When the filling velocity is higher than the critical velocity, i.e. $v>v_c$, the correlation between the fill ratio and the critical velocity can be described by the following equation:

$$\delta = \left(\frac{v_c}{v}\right)^n \tag{1}$$

where n is a material-dependent exponent. The values of v_c and n can be determined experimentally from the variation of fill ratio (δ) with filling velocity v.

Gravity filling is strongly influenced by the presence of air [2,6], as deposition of powder into the die under gravity is opposed by an adverse pressure gradient induced by air trapped in the die. For many pharmaceutical powders flow occurs with detachment of irregular lumps of material, which is known as intermittent flow [3,7]. This is an undesirable flow mechanism, as it generally yields a higher scatter in powder mass deposited into the die [3]. Experiments in vacuum have been performed to assess the influence of air in die filling [7,8]. For example, Sinka et al. [8] used an experimental die filling system, consisting of a linearly moving shoe and a stationary die. The whole die filling device was enclosed in a transparent vacuum chamber. The effect of the build-up of air pressure was proven to be significant, as a large difference in critical velocities was observed between operation in air and in a vacuum. The powder flow profiles usually appear smoother and less intermittent when experiments are conducted in a vacuum. For metallic and ceramic powders it was observed that fill densities are generally higher in a vacuum [9]. Regarding the effect of filling velocity on the efficiency of die filling in a vacuum, Goh et al. [10] performed die filling experiments using lactose powder lubricated with 1% of magnesium stearate and observed that in a linearly moving shoe operating in a vacuum, powder compression in the shoe (caused by powder inertia) leads to an increase of fill ratio at higher filling velocities. In gravity die filling, however, an increase in filling velocity reduces the fill ratio.

Further experiments have been conducted by applying suction through a downward movement of a punch/piston [8,11]. In initial studies the piston was left to fall [11], and subsequently the system was upgraded to allow for detailed studies on a wider range of suction

* Corresponding author.
E-mail address: c.y.wu@surrey.ac.uk (C.Y. Wu).

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velocities [8], where it was demonstrated that suction filling significantly improves the critical velocity by creating a negative pressure gradient under the powder bed. As pointed out by Sinka et al. [8], in suction filling there is no pre-existing air in the die, since the filling starts with the piston withdrawal only after the piston is covered by the powder in the shoe. The pressure gradient facilitates the powder flow into the die, allowing for a more efficient die filling at higher filling velocities. The authors also noticed that above the critical velocity the decrease rate of fill ratio with filling velocity (represented by n) was less abrupt than in gravity filling.

The effects of particle size and density on suction filling were investigated by Mills et al. [4], who used several grades of microcrystalline cellulose (MCC) that differed in particle size and bulk density. Due to the reduced rate of the decrease of fill ratio with the filling velocity (above the critical velocity) when suction filling is applied, the authors suggested that Eq. (1) may not be the most suitable form to describe suction filling behaviour. However, no alternative equations were proposed. Moreover, it was confirmed that suction filling yields a significantly higher critical velocity than gravity filling for all powders, although this is particularly evident for fine powders: in gravity filling the critical velocity of cohesive powders is generally very low [12,13], whereas in suction filling, their die fill performance is quite similar to that of coarse materials. The die filling of coarse powders benefits as well from the suction mechanism, however the critical velocity increase of coarse materials in suction filling compared to gravity filling is much smaller than that of fine powders.

The above-mentioned investigations focused primarily on the influence of powder properties and process parameters on die filling at high velocities (i.e. above the critical velocity) for a very limited range of powders. Very little attention was dedicated to investigating the die filling dynamics at low filling-to-suction velocity ratios, and how powder flow properties affect suction filling in these conditions. Hence, the present study aims to provide a thorough understanding of suction filling mechanisms through a detailed investigation of the combined effect of suction and filling velocity on die filling behaviour of various powders (i.e. cohesive and free flowing) over a wide range of filling velocities.

2. Materials and methods

2.1. Materials

Common pharmaceutical materials were used in this study: lactose monohydrate (Granulac® 140; Meggle, Germany), MCC Avicel PH102 (FMC; Biopolymer, Cork, Ireland), microcrystalline cellulose spheres, referred to as MCC CP102 (CELPHERE® CP102; Asahi Kasei, Japan), and Acetylsalicylic acid, referred to as ASA (Sigma-Aldrich, USA).

2.2. Particle size and morphology analysis

The particle size and sphericity distribution were analysed using the QICPIC dynamic image analysis device equipped with the GRADIS air dispersion system (Sympatec, Inc., Clausthal-Zellerfeld, Germany). The values of particle size are represented by equivalent projection circle (EQPC, defined as the diameter of a circle that has the same area as the projected area of the particle). The analyses were run in triplicate for each powder sample. Other parameters considered were span and sphericity. The span, ψ, is defined as a measure of the EQPC distribution width and is calculated as follows:

$$\psi = \frac{d_{90} - d_{10}}{d_{50}} \tag{2}$$

Sphericity is the ratio of the perimeter of the circle with the same projected area as the particle to the measured perimeter of the projected particle.

The particle morphology was assessed using scanning electron microscopy (SEM). The samples were prepared by immobilizing a small amount of each powder on a sample holder, covered in a thin layer of conductive graphite paint. The samples were subsequently gold-coated with a thickness of 4 nm. The SEM images were taken using a Jeol 7100F thermal field emission electron microscope (Jeol Ltd., Tokyo, Japan) with 3 kV accelerating voltage for lactose, MCC PH102 and MCC CP102, and a HITACHI S-3200 Scanning Electron Microscope at 5 kV for ASA, as Jeol 7100F was not available due to technical reasons.

2.3. Flow properties

The flow index was evaluated using a Flodex® flowmeter (Hanson Research Corporation, Chatsworth, USA) to assess the minimum orifice diameter that enables the powder to flow freely. The flowmeter consists of a stainless steel cylinder with a diameter of 60 mm and a height of 70 mm, with an approximate capacity of 200 cm^3, for which the base can be completed with interchangeable disks with orifices of diameter from 4 to 34 mm. The test is performed by filling the cylinder with a tested powder through a funnel to within 1 cm from the top (depending on the powder bulk density, this corresponds to 50–100 g of powder). The release lever, positioned under the cylinder, was slowly turned to open the orifice and to allow powder flow. If the hole at the bottom of the cylinder was visible after powder discharge, it indicated that the powder could flow through the given orifice. In that case the test was repeated by decreasing the orifice diameter until no flow was observed. The flow index was defined as the diameter of the smallest orifice that enabled the powder flow.

Flow properties were also collected using the semi-automated granular material flowmeter, GranuFlow (Granutools™, Belgium). GranuFlow consists of a stainless steel cylindrical silo of 47 mm diameter and 300 mm length, positioned above an aluminium rotating plate with apertures of various diameters. A precise amount of material (100 g) was poured through a funnel in the cylinder. The plate was rotated until an aperture of a certain diameter, that would enable hopper discharge, was positioned under the silo. The test was performed using apertures of 4, 6, 8, 10, 12, 14 and 16 mm. The powder discharge was registered with a weighing system, connected to data analysis software. For each aperture size, the mass flow rate (g/s) was obtained. The step was repeated for apertures of increasing size. Each measurement was carried out in triplicate. The mass flow rate of granular materials, obtained from the measurements, was plotted as a function of orifice size in order to obtain the best fit curve using the following equation:

$$W/\sqrt{g} = C_b(D - D_{min})^{\frac{5}{2}} \tag{3}$$

where W is the mass flow rate, g is the gravitational acceleration, C_b is the flowability index, dependant on the bulk density, D is the aperture diameter and D_{min} is treated as the minimum aperture diameter that allows powder to flow [14]. The values of C_b and D_{min} are obtained by fitting the plots of mass flow rate as a function of aperture size using Eq. (3), in order to compare the flow properties of analysed materials.

2.4. Air permeability and aeration properties

Dynamic flow properties of materials, such as aeration and air permeability, were evaluated using a Freeman FT4 Powder Rheometer (Freeman Technology, Tewkesbury, UK), referred to as FT4. All measurements were performed in triplicate at 23 ± 2 °C–50 ± 10% relative humidity.

The aeration test was carried out using a borosilicate vessel of 260 cm^3 and 50 mm diameter (140 mm height), equipped with a porous aeration base connected to an aeration unit. The flow resistance of the powder sample was measured under an increasing level of air velocity. Each measurement was preceded by a conditioning cycle: the

48 mm diameter blade was rotated in a clockwise direction as it was driven downward into the bed at a helix angle of 5° and a tip speed of 60 mm/s until the blade was 10 mm from the base of the vessel, before reversing the direction of vertical movement. The air velocity was then set for the test cycle and the impeller was driven downwards with anti-clockwise rotation, a helix angle of 5° and a tip speed of 100 mm/s until the blade was 10 mm from the base of the vessel, before reversing the direction of vertical and rotational movement. Air velocities of 0–10 mm/s were used in the test cycles, with an increment of 2 mm/s. The flow energy, the cumulative rotational and vertical energy required to drive the impeller downward into the bed, was registered for each air velocity. The air velocity was adjusted for all the materials in order to achieve a stabilised value of flow energy.

The air permeability test was carried out using a borosilicate split vessel of 85 cm^3, equipped with aeration base connected to the aeration unit, a 48 mm vented piston and 48 mm conditioning blade. The test was preceded by three conditioning cycles, identical to the conditioning cycle of the aeration test. After the conditioning step, the blade was replaced by the vented piston. The powder bed was subjected to increasing values of normal pressure, ranging from 1 to 15 kPa. Each normal stress was applied for 60 s, while at the same time air was passed through the powder bed at 2 mm/s for all the powders considered and additionally at 10 mm/s for MCC PH102, MCC CP102 and ASA. The pressure drop, resulting from the pressure difference between the top and the bottom of the container, was then recorded. Generally, a higher pressure drop at a given normal pressure is generated by finer and more cohesive powders that tend to produce a tightly packed structure [15,16].

2.5. Die filling

A suction die filling system was developed (see Fig. 1), which consisted of a pneumatically driven cuboidal feed shoe, moving linearly over a stationary die. The shoe has inner dimensions of 150 mm length, 30 mm depth and 55 mm height. The shoe is equipped with a perforated, removable rectangular lid that prevents powder escaping from the shoe as the shoe is accelerated to high velocities. The velocity of the shoe during the filling step, known as filling velocity, can be controlled through a dedicated control unit. The die is made of aluminium, with a transparent plexiglass front wall of 12 mm thickness. The die opening is a square of 20 mm per side. The maximum filling depth was 50 mm and could be adjusted though the vertical displacement of a brass piston. The piston was connected to an electromechanical toothed belt axis (Festo Ltd., UK), which can reach a linear velocity of 3 m/s and a maximum acceleration of 30 m/s^2. In order to ensure that the filling of the die is performed only when it is covered, the piston withdrawal starts as soon as the leading edge of the shoe completely covers the die opening. The investigated filling and suction velocity ranges were 50–800 and 50–200 mm/s, respectively. The efficiency of die filling was assessed in terms of fill ratio, σ, which was determined by:

$$\sigma = \frac{m}{M} \tag{4}$$

where m is the mass of powder in the die at a given filling velocity, and M is the mass of powder in a fully filled die. Both masses were measured using a Kern PCB1002 laboratory balance with a sensitivity of 0.01 g (Kern&Sohn GmbH, Germany). The powder flow behaviour was also captured using a Phantom v1612 high speed camera at 1000 fps (Phantom VEO, Vision Research Inc., USA).

Since die filling occurs as a result of two simultaneous movements, of the shoe and the piston, there is an optimal range of velocities that should theoretically account for the maximum filling, as explained in detail by Wu et al. [12]. The suction starts when the die is covered by the leading edge of the shoe and continues during the shoe transit

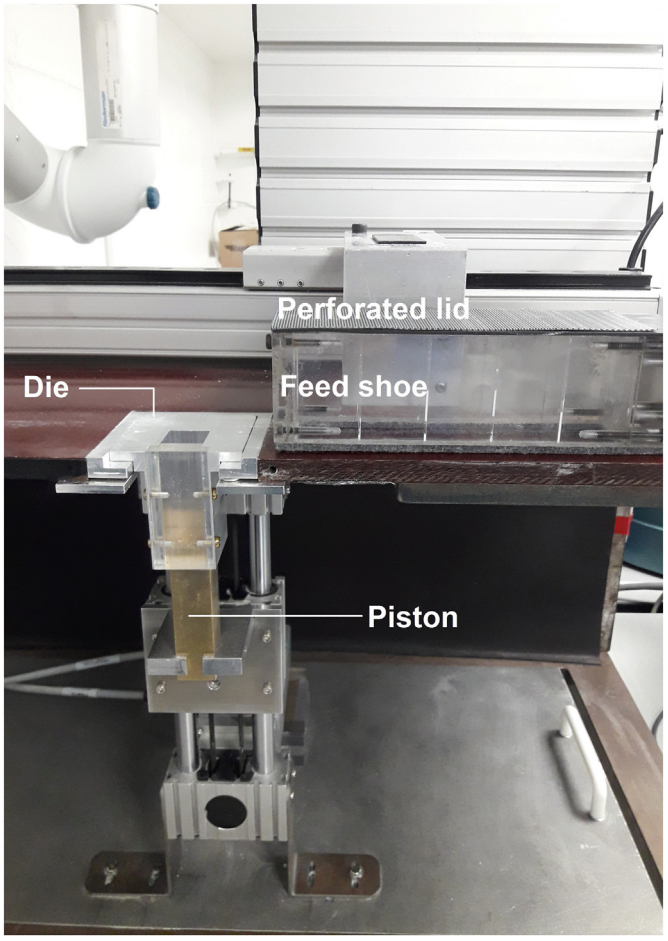


Fig. 1. A model suction die filling system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

over the die. As the trailing edge of the shoe travels across the die, the effective discharge area becomes smaller than the die opening, since the area of the die covered by the shoe becomes increasingly smaller. At this point the suction is less efficient, as the die is partially exposed to air. Therefore, for a given suction velocity, there is an *optimal filling velocity*, v_o, below which the piston completes the specified displacement before the die is exposed to the air again. The optimal filling velocity is determined as follows:

$$v_o = \frac{L_s - L_d}{H / v_p} \tag{5}$$

where L_s is the shoe length, L_d is the die width, H is the filling height (also equivalent to the maximum piston displacement) and v_p is the suction velocity (where p stands for "piston"). Eq. (5) can be rewritten as:

$$\frac{v_o}{v_p} = \frac{L_s - L_d}{H} \tag{6}$$

where v_o is the optimal filling velocity. Theoretically, if the filling velocity is below or equal to the optimal filling velocity, the fill ratio should always be equal to 1.0, or as close as possible to this value, and therefore independent of the filling velocity. When the optimal filling velocity is exceeded, the piston still moves downwards even when there is no powder above the die. Therefore only a fraction of the die volume is available to be filled whilst there is powder above the die, with this fraction decreasing as filling velocity is increased (for a given suction

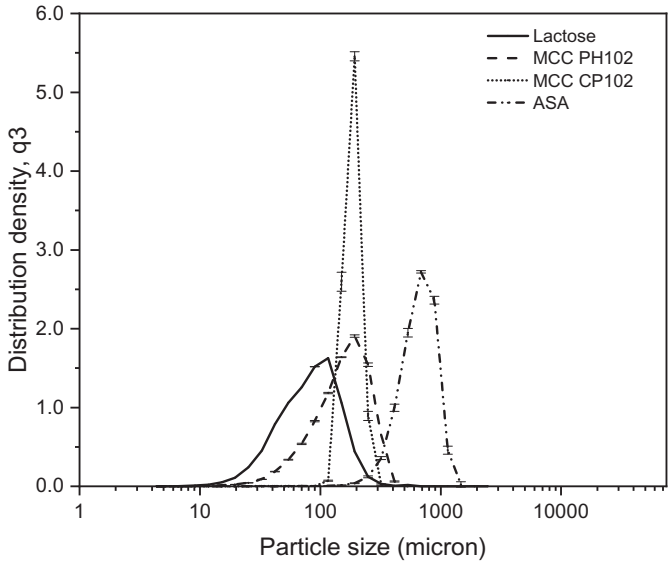


Fig. 2. Particle size distributions for all powders considered.

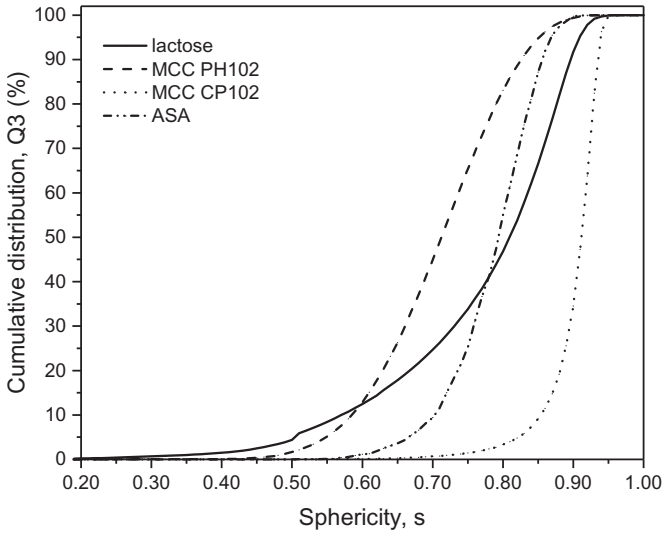


Fig. 3. Cumulative sphericity distributions for all powders considered.

velocity). As such the fill ratio decreases as the filling velocity increases, for a given suction velocity.

3. Results and discussion

3.1. Particle size and morphology

The particle size distribution of the examined materials is given in Fig. 2. It can be seen that lactose and MCC PH102 are fine powders with the largest spans, whereas MCC CP102 and ASA consist of coarser particles with narrower spans. The average values of particle size are presented in Table 1, along with the sphericity. From the span values it is evident that Lactose and MCC PH102 have wider dispersions of particle size than MCC CP102 and ASA. In terms of sphericity, the material that has the most spherical particles is MCC CP102, which also appears to have the narrowest sphericity distribution. MCC PH102 and ASA appear similar in terms of sphericity values, whereas lactose consists of the least spherical particles. This is confirmed by the cumulative sphericity distribution, shown in Fig. 3.

From the SEM images (Fig. 4) it can be seen that lactose (Fig. 4a) mostly contains irregular particles as well as large aggregates. In case of MCC PH102 (Fig. 4b) coarser and more regular particles coexist with smaller, fibrous-like fragments. MCC CP102 (Fig. 4c) consists of particles of slightly varying size but of overall highly spherical shape. Finally, ASA (Fig. 4d) consists of oblong crystals and smaller particles derived from the material fragmentation.

3.2. Flow properties

The flow index values are reported in the first column of Table 2. The flow index value for lactose is the highest, followed by MCC PH102, then MCC CP102 and ASA, both of which have a flow index of 5 mm. Their flow behaviour can be explained in terms of particle size and shape. It is well known that the increase of particle size is accompanied by the decrease of cohesion [17]. Lactose consists of fine and the most irregularly shaped particles (as shown in Fig. 4a). MCC PH102 particles are coarser, but in terms of shape they are irregular as well (Fig. 4b). ASA has a low flow index because it consists of very large particles (Table 1), whereas MCC CP102 consists of particles similar in size to the less free flowing MCC PH102 (Table 1), however its flowability is enhanced by its highly spherical particle shape (Fig. 4c).

The mass flow rate values, obtained from GranuFlow, are plotted as a function of aperture diameter in Fig. 5. Lactose is excluded from the figure since no flow was achieved. The mass flow rate increase with aperture size is more significant for MCC CP102 and it is the lowest for MCC PH102. The mass flow rate of ASA is positioned between MCC CP102 and MCC PH102 for all aperture diameters. MCC PH102 presents higher scatter, indicating less constant and reproducible flow.

The flowability index, C_b, and minimum diameter, D_{min}, obtained from the mass flow rate data, are reported in Table 2. MCC CP102 has the lowest C_b, whereas ASA and MCC PH102 both have greater values of C_b, with the latter being greatest. The minimum orifice diameter necessary for a flow to start (D_{min}), is the lowest for MCC CP102 and the highest for MCC PH102, with ASA having a value in between. No values of C_b or D_{min} were obtained for lactose as it was impossible to achieve any flow using GranuFlow. Since high values of C_b and low values of D_{min} are linked to better flowability [14], MCC CP102 is the most free-flowing material and MCC PH102 has the poorest flowability (excluding lactose). According to this data, the powder with the highest flowability is ASA, whereas MCC PH102 has the poorest flow properties (lactose is not included). These results are in accordance with flow index values, shown in Table 2: ASA and MCC CP102 have the lowest flow index value (therefore the highest flowability), whereas MCC PH102 has a significantly higher flow index (lower flowability).

Table 1
Size parameters data from QICPIC Particle Size Analyser (data expressed as mean ± standard deviation).

Powder	EQPC				Sphericity		
	d_{10} (μm)	d_{50} (μm)	d_{90} (μm)	span (-)	s_{10} (-)	s_{50} (-)	s_{90} (-)
Lactose monohydrate	36.25 ± 0.62	84.95 ± 2.23	159.28 ± 6.32	1.46 ± 0.09	0.600 ± 0.042	0.648 ± 0.012	0.807 ± 0.005
MCC PH102	68.24 ± 0.38	161.60 ± 0.81	274.89 ± 0.43	1.28 ± 0.01	0.584 ± 0.002	0.712 ± 0.002	0.825 ± 0.001
MCC CP102	143.84 ± 0.59	186.60 ± 0.95	221.40 ± 2.14	0.42 ± 0.01	0.851 ± 0.003	0.912 ± 0.001	0. 928 ± 0.001
ASA	402.57 4.56	672.64 7.37	966.51 8.49	0.44 ± 0.02	0.577 ± 0.164	0.692 ± 0.141	0.808 ± 0.068

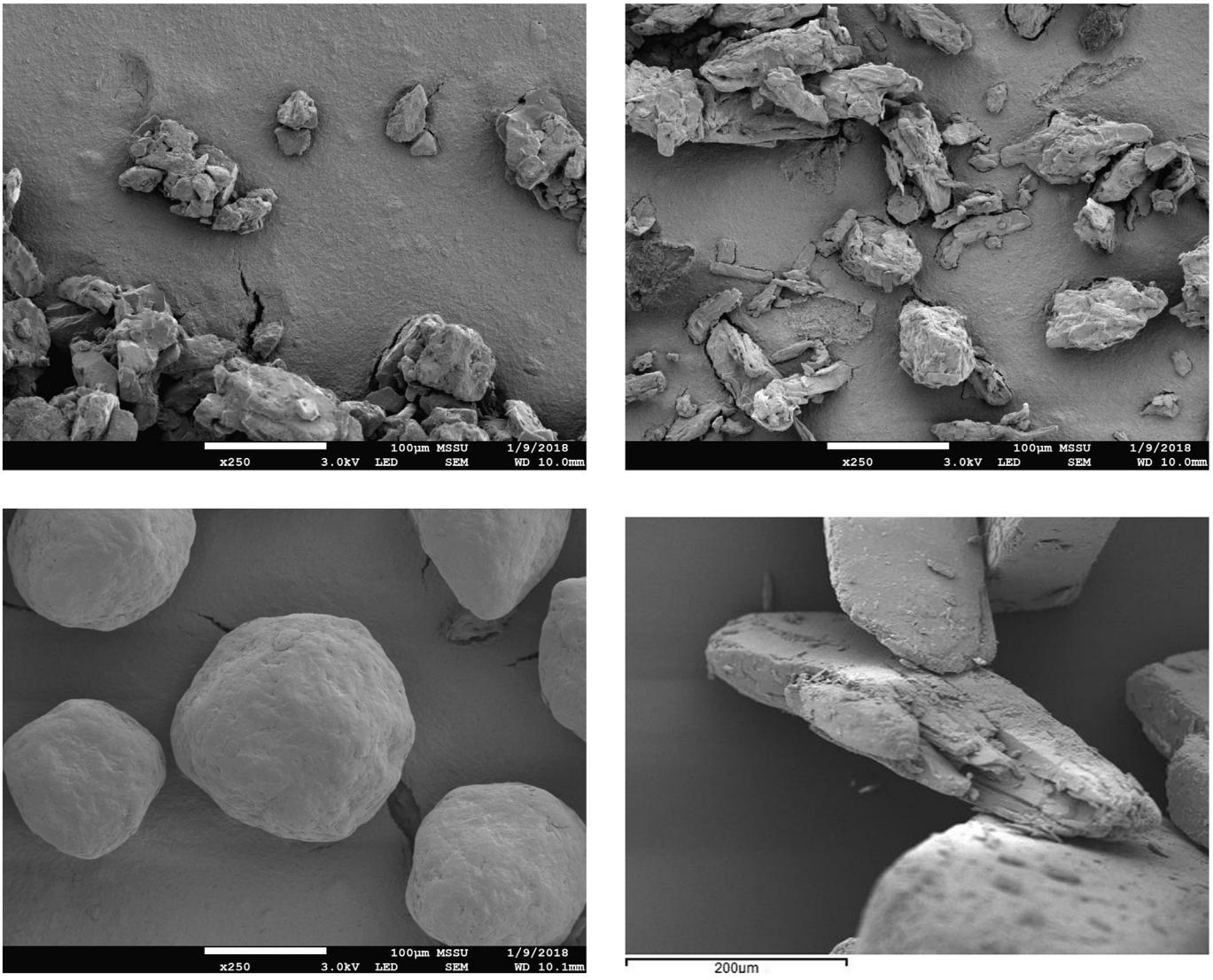


Fig. 4. a) Lactose monohydrate. b) MCC PH102. c) MCC CP102. d) ASA. Scanning electron microscope images of all powders considered.

3.3. Air permeability and aeration properties

The powder behaviour under aeration is presented in Fig. 6, where the flow energy is plotted as a function of air velocity. It can be seen that the aeration curve of ASA is significantly different from other powders: at the same air velocity the associated flow energy is higher than that of the other three powders. The fluidisation curves of the other three materials can be seen in more detail in the top right plot. For MCC CP102 fluidisation occurs at approximately 15 mm/s (signified by a flow energy of zero) and the flow energy decreases almost linearly with the increase of air velocity below the fluidisation velocity. MCC PH102 shows a sharp drop in flow energy as the air velocity increases, followed by a more gradual drop around 6–10 mm/s, with full fluidisation occurring at 10 mm/s. Lactose shows similar values of flow energy

Table 2
Flow index, C_b and D_{min} calculated for the examined powders (data expressed as mean ± standard deviation).

Powder	Flow index (mm)	C_b (g/mm^3)	D_{min} (mm)
MCC PH102	12	$(2.8 \pm 1.2)x10^{-4}$	5.0 ± 1.1
MCC CP102	5	$(5.9 \pm 0.3)x10^{-4}$	0.7 ± 0.1
ASA	5	$(4.6 \pm 0.1)x10^{-4}$	1.5 ± 0.1
Lactose monohydrate	24	–	–

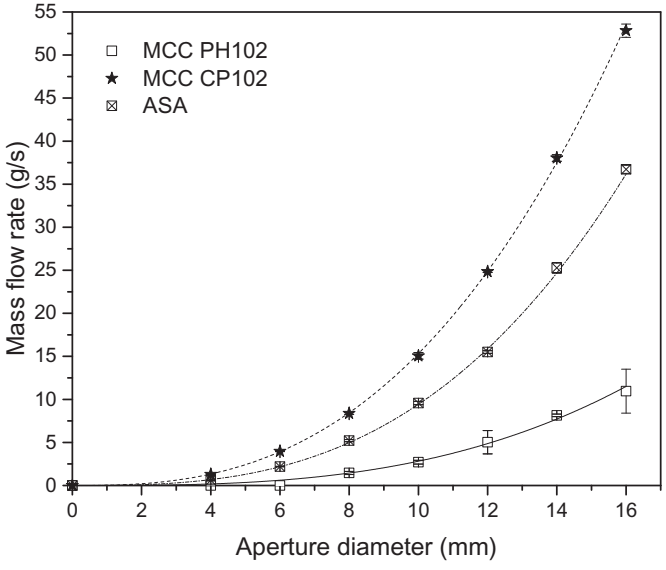


Fig. 5. Mass flow rate profiles of MCC PH102, MCC CP102 and ASA.

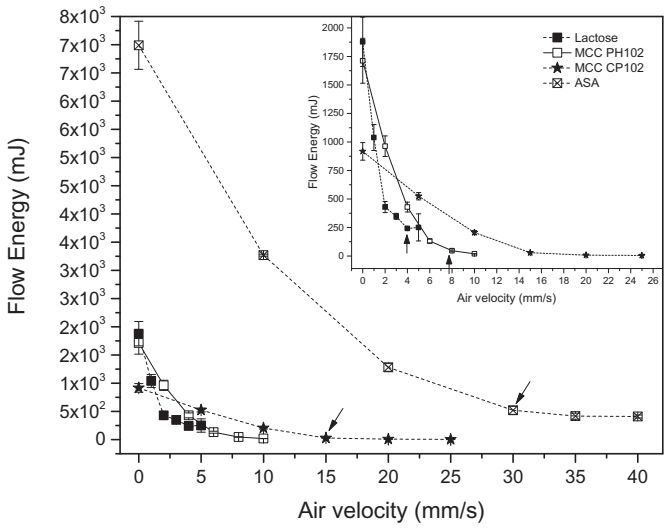


Fig. 6. Flow energy as a function of air velocity for all powders considered. Black arrows indicate the onset of fluidisation.

to MCC PH102 in absence of fluidisation, but at low air velocities the flow energy decreases more sharply than MCC PH102, and lactose seems to be fluidised at around 5 mm/s.

In order to explain this behaviour, it is necessary to consider that a powder becomes fluidised when the air flow efficiently separates its particles, thus counter balancing both the interparticle interaction and the gravity force. Although a more irregularly shaped particle should disrupt the air flow, creating turbulence that aids fluidisation, in the case of the highly irregular ASA particles the fluidisation still occurs at high velocities. This could be justified by the particle size, since coarser and heavier particles could be more difficult to suspend than lighter and smaller particles of more regular shape [18]. This is the reason why MCC CP102, which consists of spherical particles with optimal aerodynamic properties, requires lower velocities for its fluidisation than ASA. Irregular particle shape of MCC PH102 (Fig. 4b) and smaller size (if compared to MCC CP102) are most likely the reason of lower fluidisation velocity of MCC PH102 compared to MCC CP102.

A sharp drop in the flow energy of lactose and MCC PH102 is most likely due to their lower permeability, as shown by Leturia et al. [19]. The irregular shape of these particles results in more tightly packed

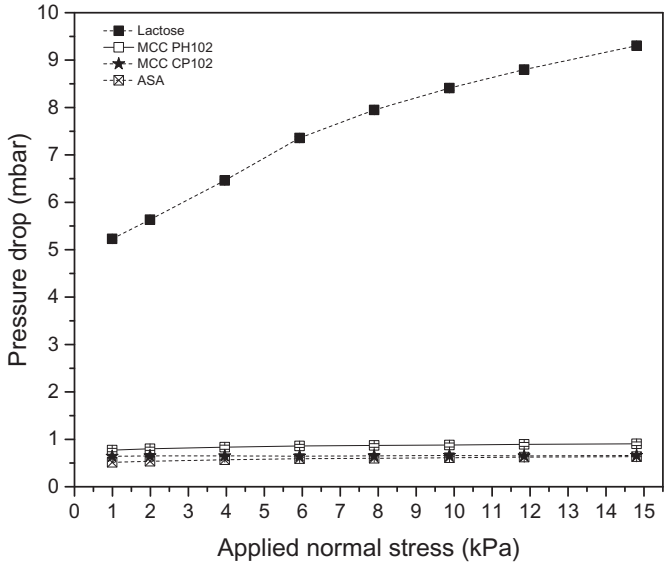


Fig. 7. Pressure drop across the powder bed at an air velocity of 2 mm/s for the powders considered.

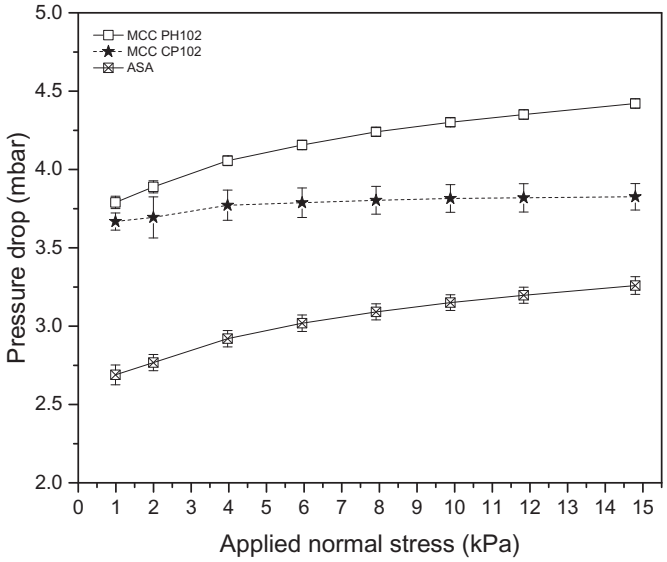


Fig. 8. Pressure drop across the powder bed at an air velocity of 10 mm/s for MCC PH102, MCC CP102 and ASA.

beds that account for a higher initial flow energy. As the air penetrates in the spaces between small and light particles, a slight increase in the air velocity results in efficient separation of the particles, which quickly become fluidized.

The plots of pressure drop (indicating permeability) under the application of normal stress in the presence of a 2 mm/s upward air velocity are presented in Fig. 7. It is evident that lactose exhibits the highest pressure drop at all consolidation levels considered, as observed by Fu et al. [15] for lactose grades of a comparable particle size, and for this material the pressure drop increases under the application of normal stress. Compared to lactose, the other three materials present values of pressure drop close to zero and are therefore highly permeable. In order to obtain a more significant difference between the pressure drop values for these powders, it is necessary to examine them at higher air velocities. Therefore, in Fig. 8 the pressure drop curves of MCC PH102, MCC CP 102 and ASA measured at an air velocity of 10 mm/s are presented. MCC PH102 and MCC CP102 present similar permeability at the lowest stress, but as the applied normal stress increases the pressure drop across the bed of MCC PH102 increases more than that of MCC CP102. The lowest pressure drop (and therefore highest initial permeability) is observed for ASA. The pressure drop increases as the powder bed permeability decreases under the effect of particle rearrangement and increased packing density as a result of applied normal stress. Fine, cohesive materials, like lactose, generally experience a greater increase in packing fraction with applied stress, which causes the air permeability to decrease significantly. Coarser particles offer less surface area for interlocking, and generally the powder bed porosity remains higher, allowing air to pass through the consolidated bed [19,20]. Therefore greater air velocities are required to produce a given pressure drop between the top and the bottom of the powder bed. This is the case for MCC PH 102, MCC CP102 and ASA. Despite quite similar particle sizes (see Table 1), MCC PH102 and MCC CP 102 present different permeability behaviour, since it is influenced by the packing structure [20], which depends on the particles shape, which is more irregular for MCC PH102. The spherical shape of MCC CP102 microspheres provides a more regular packing structure, which limits the change in porosity in the range of applied normal stresses. For ASA the change in pressure drop with applied normal stress is very similar to that of MCC PH102, although the coarse nature of the particles results in a greater permeability, and therefore lower pressure drop.

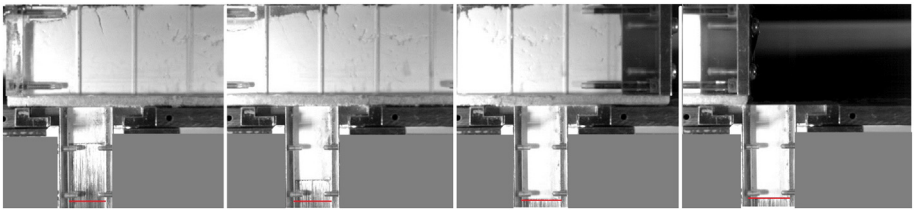


Fig. 9. Powder flow profiles of lactose monohydrate at a suction velocity of 100 mm/s and a filling velocity of 100 mm/s ($v_r < v^*$). The red line indicates the specified terminal position of the piston. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

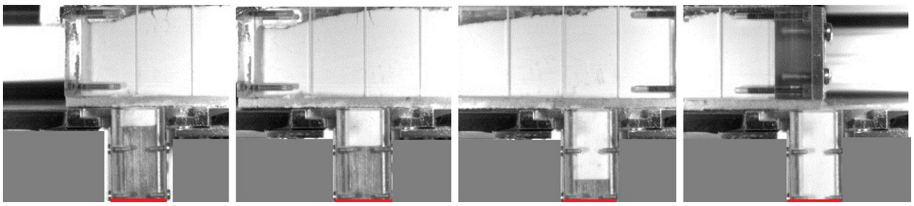


Fig. 10. Powder flow profiles of lactose monohydrate at a suction velocity of 100 mm/s and a filling velocity of 260 mm/s ($v_r = v^*$). The red line indicates the specified terminal position of the piston. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

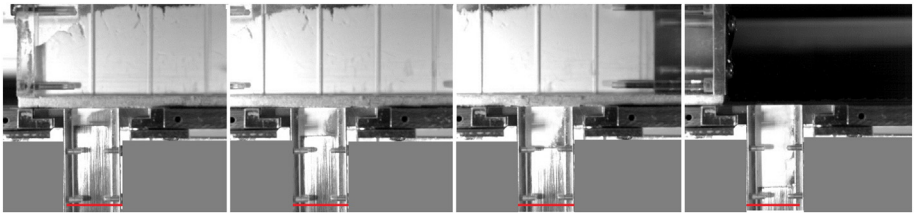


Fig. 11. Powder flow profiles of lactose monohydrate at a suction velocity of 100 mm/s and a filling velocity of 500 mm/s ($v_r > v^*$). The red line indicates the specified terminal position of the piston. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

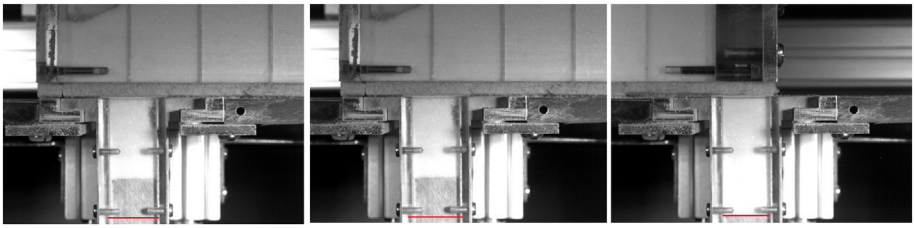


Fig. 12. Powder flow profiles of ASA at a suction velocity of 200 mm/s and a filling velocity of 160 mm/s ($v_r < v^*$). The red line indicates the specified terminal position of the piston. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

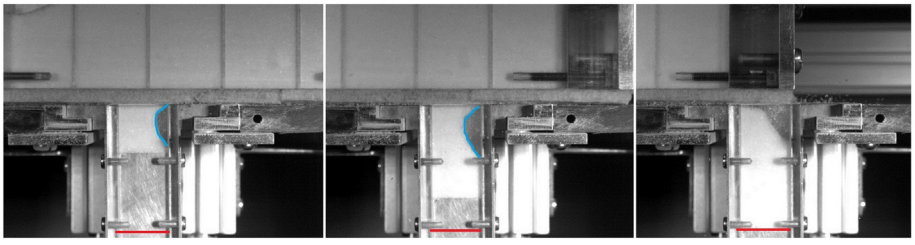


Fig. 13. Powder flow profiles of ASA at a suction velocity of 200 mm/s and a filling velocity of 520 mm/s ($v_r = v^*$). The red line indicates the specified terminal position of the piston. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

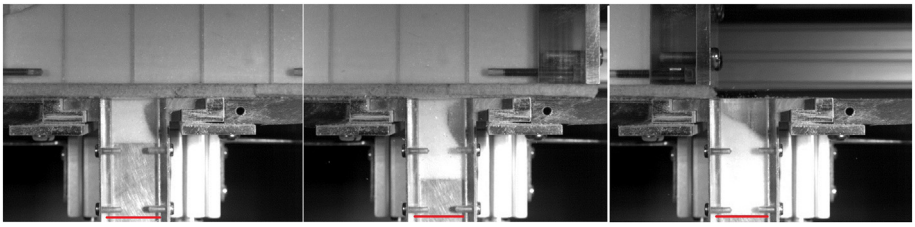


Fig. 14. Powder flow profiles of ASA at a suction velocity of 200 mm/s and a filling velocity of 600 mm/s. The red line indicates the specified terminal position of the piston. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.4. Die filling

3.4.1. Powder flow profile

The flow profile images for lactose are shown in Figs. 9–11. Depending on the suction and filling velocities (the filling to suction velocity ratio v_r), the die filling process could be divided into three characteristic regimes:

1. In slow filling ($v_r < v^*$), when a high suction velocity is combined with a low filling velocity, intermittent flow is observed (Fig. 9) and the die is not completely filled.
2. Plug flow at synchronised filling, where $v_r = v^*$ (Fig. 10). The powder experiences plug flow, since it is drawn inside the die as a continuous column of material [4,12] and the die is completely filled.
3. Plug flow, albeit incomplete, in fast filling, where $v_r > v^*$ (Fig. 11). The powder is drawn inside the die as a disrupted column of material, and the die is not filled completely at the end of this step, since the piston does not reach the terminal position before the shoe transition is complete.

The observed increase of die filling efficiency of lactose with increasing filling velocity (with the maximum fill ratio reached at synchronised filling) can be explained in terms of powder densification, associated with the initial shoe acceleration, as explained by Wu et al. [12] and experimentally observed by Goh et al. [10] in vacuum filling. As the powder moves in the shoe across the die it is also slightly compacted towards the trailing edge of the shoe, due to powder bed inertia and friction with the die table surface when the shoe is accelerating from stationary to a specified filling speed. The densification degree increases as the filling velocity increases. This leads to an improvement of die filling efficiency as the filling velocity increases, until the optimal velocity is reached (i.e. synchronised filling takes place). At that point, the compacted column of powder is efficiently drawn inside the die as a plug (Fig. 10). In fast filling, the powder compaction increases further due to its inertia at a high filling velocity, whereas the suction velocity remains the same. As a consequence, the produced suction is less efficient at drawing the whole column of powder into the die. This results in an initial nose-like flow, followed by a plug flow, once the powder has started to fall into the die (Fig. 11).

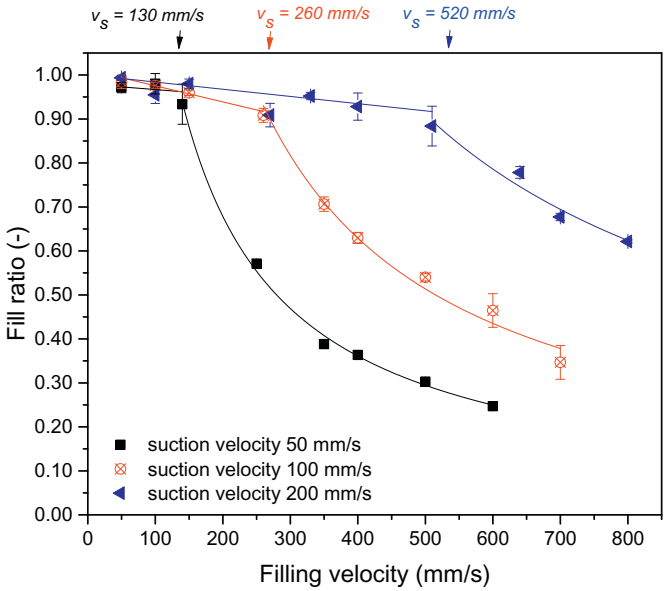


Fig. 16. The fill ratio as a function of the filling velocity for MCC PH102 at various suction velocities. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

MCC PH102 and MCC CP102 both show plug-like flow at all filling velocities. However, the die filling profile of ASA appears to be different. In slow filling, the flow is also plug-like (Fig. 12), whereas at synchronised filling the flow is not continuous and the powder is not drawn in the die as a compacted column (Fig. 13). The piston withdrawal creates an apparent nose flow (Fig. 13, highlighted in blue). This type of flow, described in detail by Wu et al. [1,2,4], occurs in gravity filling as the shoe starts its transition over the die and a stream of powder, falling from the front region of the powder bed, builds up in the leading corner. What appears to be a nose flow for ASA is an incomplete plug flow, caused by a less efficient densification of the material, which leads to an incomplete filling of the die. In fast filling, where the filling velocity becomes higher than the optimal velocity (Fig. 14), the apparent "nose" flow becomes more evident.

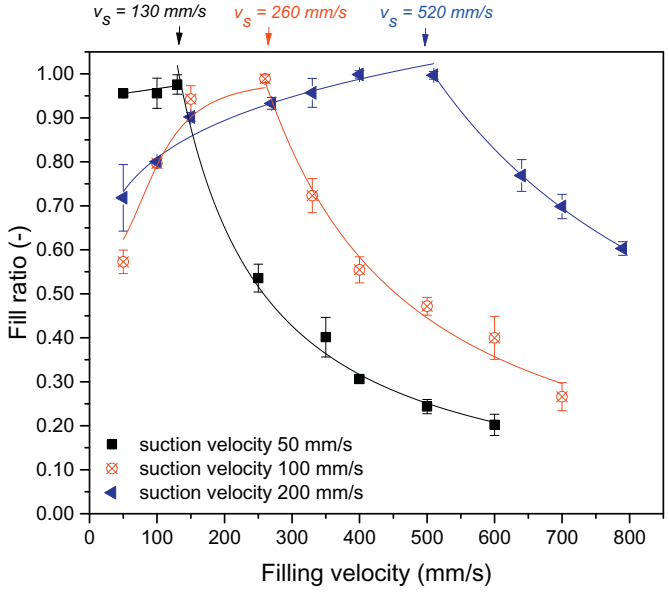


Fig. 15. The fill ratio as a function of the filling velocity for lactose at various suction velocities. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

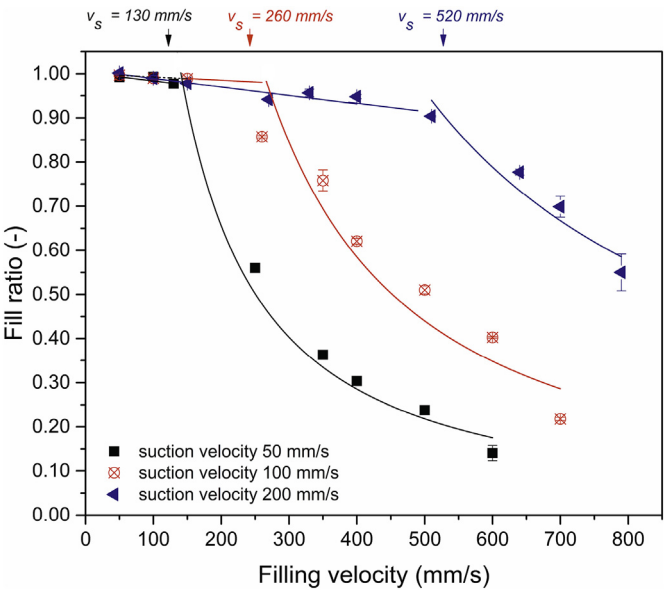


Fig. 17. The fill ratio as a function of filling velocity for MCC CP102 at various suction velocities. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

An efficient plug flow is achieved for cohesive powders that are compacted more efficiently and have lower air permeability. This is why suction filling was shown to be more efficient for fine materials [4]. Observing the drop in air permeability under the effect of normal stress (Figs. 7, 8) it can be seen that lactose yields a pressure drop an order of magnitude greater than the other powders. Therefore, lactose produces the most compacted plug of material, which, in turn, explains a greater sensitivity of lactose die filling efficiency to the filling velocity: as the shoe is accelerated to higher velocities, the materials undergoes a more significant compaction. When the compacted bed is subjected to suction, the filling occurs through a deposition of a continuous plug into the die, due to lactose lower air permeability. On the other hand, the very coarse ASA forms the most permeable bed and its mechanism of filling is mostly unaffected by the filling velocity.

3.4.2. Die filling efficiency

The die filling profiles of lactose at different suction velocities are presented in Fig. 15. At a suction velocity of 50 mm/s (black curve), the fill ratio increases from approximately 0.95 at a filling velocity of 50 mm/s, to 1.00 at a filling velocity of 130 mm/s (v^*). Theoretically it would be expected that a fill ratio of 1.00 could be achieved for any filling velocity equal to or below v^*, however the intermittent flow occurring in slow filling (where $v_r<v^*$), described in Section 3.4.1, prevents this for lactose. Beyond the filling velocity of 130 mm/s, the fill ratio decreases exponentially as fast filling occurs.

For the suction velocity of 100 mm/s, the corresponding optimal filling velocity (v^*) is 260 mm/s (synchronised filling). Hence slow and fast filling occur respectively below and above 260 mm/s. So, at a suction velocity of 100 mm/s (red curve), the fill ratio at 50 mm/s filling velocity is around 0.55, and it reaches the maximum value of 1.0 at a filling velocity of 260 mm/s (Fig. 10). In fast filling (Fig. 11), the fill ratio decreases exponentially, and the die is never filled completely.

Finally, the suction velocity of 200 mm/s produces synchronised filling at 520 mm/s of filling velocity. Therefore, at a suction velocity of 200 mm/s (blue curve), the fill ratio increases from 0.70 at a filling velocity of 50 mm/s (slow filling) to 1.0 at the optimal filling velocity of 520 mm/s (synchronised filling), followed by an exponential decrease in fast filling.

The fill ratio profiles of MCC PH102 are presented in Fig. 16. It can be seen that for each suction velocity the fill ratio in slow filling tends to decrease slightly as the optimal filling velocity, v^*, is approached. So, in slow filling at a suction velocity of 50 mm/s (black curve) the fill ratio gradually drops from 1.0 to approximately 0.95. At a suction velocity of 100 mm/s (red curve) the fill ratio drops from 1.0 to slightly above 0.9 when synchronised filling is reached. At a suction velocity of 200 mm/s (blue curve), the fill ratio reaches 0.9 in synchronised filling. In all cases the fill ratio decreases exponentially in fast filling. The fill ratio profiles of MCC CP102 are shown in Fig. 17. It can be seen that the fill ratio profiles at all suction velocities considered look very similar to that of MCC PH102. The same can be said about ASA fill ratio profiles, presented in Fig. 18.

It is evident that for a given powder, all three fill ratio profiles, obtained at various suction velocities, are similar and therefore can potentially be represented as a single curve. In order to obtain such a plot, the values of fill ratio at all the examined suction velocities need to be represented as a function of the filling to suction velocity ratio (v_r). A single master curve that includes all data of fill ratio for lactose is presented in Fig. 19. The concepts of slow, synchronised and fast filling can be applied

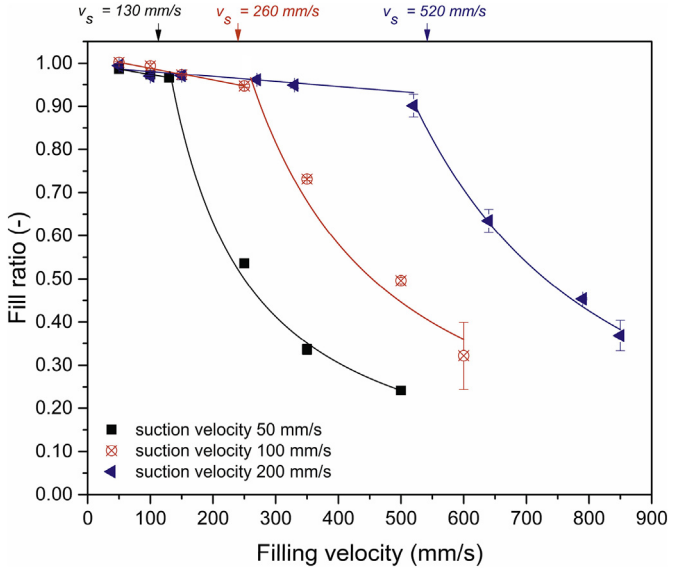


Fig. 18. The fill ratio as a function of filling velocity for ASA at various suction velocities. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

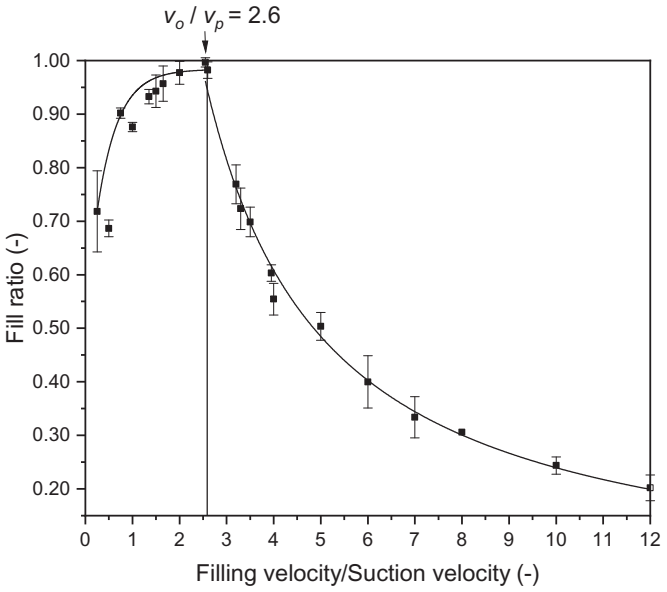


Fig. 19. The fill ratio as a function of filling velocity to suction velocity ratio for lactose monohydrate.

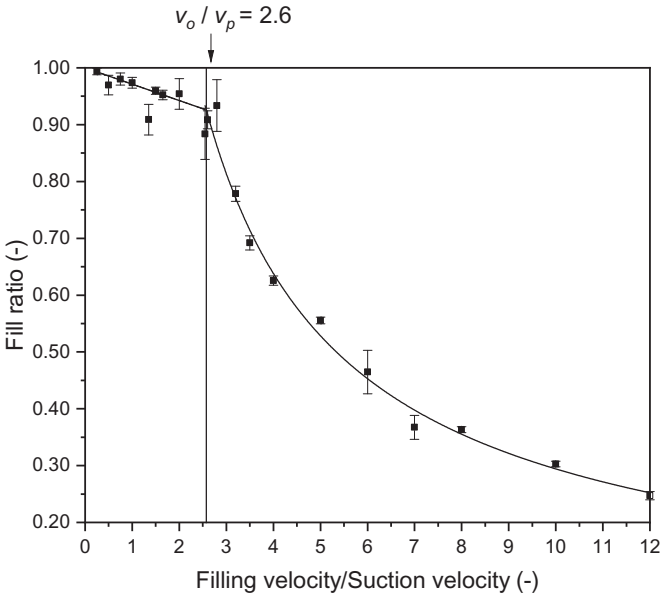


Fig. 20. The fill ratio as a function of filling velocity to suction velocity ratio for MCC PH102.

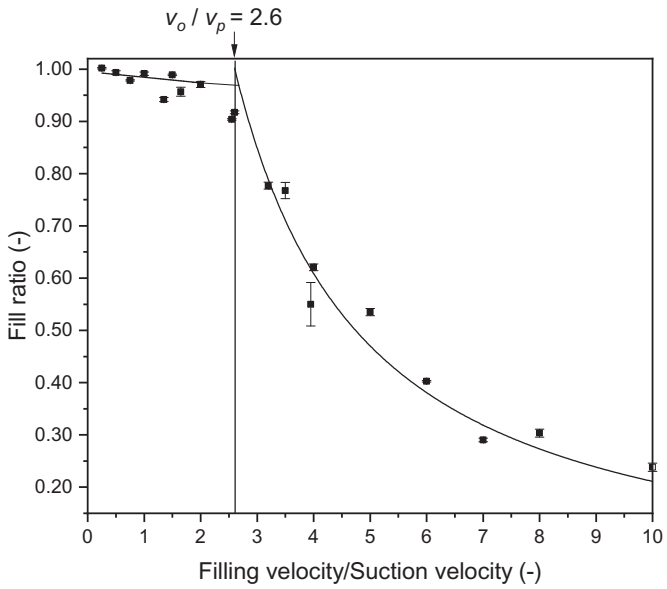


Fig. 21. The fill ratio as a function of filling velocity to suction velocity ratio for MCC CP102.

to the master curve. At all suction velocities the fill ratio of lactose tends to increase as v_r increases (slow filling), until v^* is reached, and the synchronised filling takes place. The value of v^* is given by Eq. (6), as it represents the combination of filling and suction velocity that allows the piston to complete its vertical displacement within the same time it takes the shoe to transit over the die. The value of v^* is constant (equal to 2.6 for the system used in the study), unless such factors as filling depth, shoe length or die width are modified.

Above v^* (fast filling) the fill ratio decreases, as the shoe completes its pass over the die before the piston reaches the bottom position. In contrast, the free flowing MCC PH102 (Fig. 20), MCC CP102 (Fig. 21) and ASA (Fig. 22) show a gradual decrease in fill ratio in slow filling at increasing v_r.

Observing the master curves of MCC PH102, MCC CP102 and ASA, it is evident that they all share a similar profile and therefore can all coalesce into a single higher-level master curve, consisting of a linearly decreasing section followed by the exponentially decreasing section. The

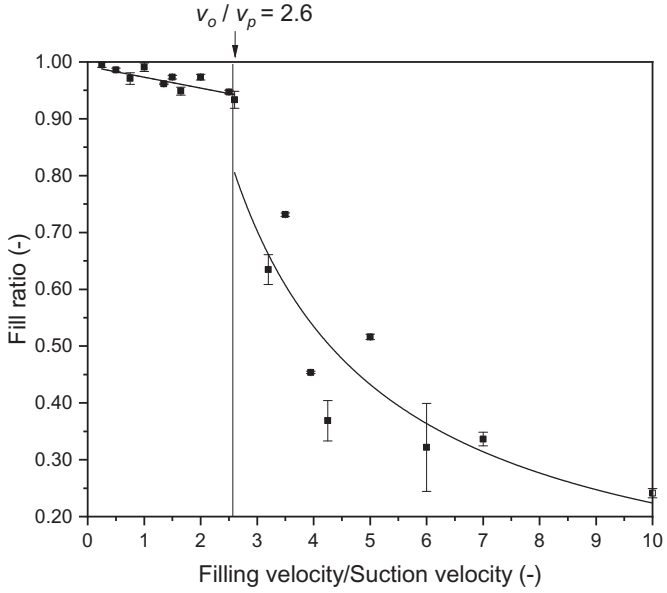


Fig. 22. The fill ratio as a function of filling velocity to suction velocity ratio for ASA.

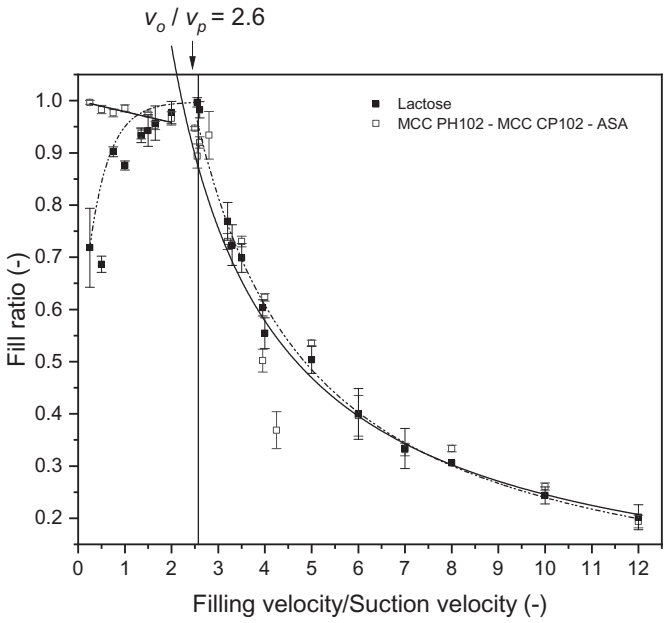


Fig. 23. Fill ratio as a function of filling velocity/suction velocity for the lactose (continuous line) and MCC PH102, MCC CP102 and ASA (dashed line).

comparison between the (cohesive) lactose master curve and the free flowing materials master curve is shown in Fig. 23.

A possible explanation for the difference in slow filling observed for two types of powders could be a lower suction efficiency, resulting from a higher air permeability (Figs. 7, 8). Another reason for the fill ratio decrease observed for coarse materials could be that coarser and heavier particles have greater inertia. Therefore, as they are drawn into the die by suction, these particles maintain the horizontal velocity caused by the shoe movement, and form a nose-like profile. This is particularly evident for coarser and heavier particles of ASA (Fig. 13, the nose is highlighted in blue). The resulting amount of material flowing in the die tends to decrease, as this nose-like flow provides a smaller volume of powder than a continuous plug flow. In fast filling both cohesive and free flowing materials undergo the same exponential decrease of fill ratio, which is evident from observing both the fill ratio vs filling velocity (Figs. 15–18) and fill ratio vs filling to suction velocity profiles (Fig. 23). This uniformity in filling behaviour observed for all types of powders comes as a result of plug flow being the predominant mechanism: throughout the whole filling process in fast filling the powder is subjected to continuous suction (since the piston withdrawal continues even after the shoe has transited over the die). On the contrary, in slow filling, the piston withdrawal is completed before the shoe has transited over the die, therefore the suction filling is inevitably accompanied by gravity filling as well, where the powder properties play an important part in influencing the die filling. Therefore, in slow filling, differences can be observed between various types of powders, whereas in fast filling these differences play an insignificant role.

4. Conclusions

Suction die filling behaviours of four typical pharmaceutical powders were investigated. All materials differed in terms of particle size, shape, flowability and air permeability. It was observed that for the fine, cohesive lactose powder, the die filling behaviour is influenced not only by the suction velocity, but also by the filling velocity, and there is a combination of suction velocity and filling velocity that yields the maximum die filling efficiency: the higher the suction velocity, the higher the corresponding filling velocity that produces the maximum fill ratio.

For a given suction filling system an optimal filling to suction ratio (v^*) was identified. The fill ratio of cohesive materials was low at low v_r, but it showed an exponential increase at increasing v_r, reaching the maximum value at v^*. This is because compaction increases and permeability decreases for cohesive powders, as they are accelerated to higher velocities that allow the powder to be drawn into the die as a continuous plug of material.

Coarse and more freely flowing materials, on the other hand, showed the maximum fill ratio at low v_r. Fill ratio was observed to decrease at increasing v_r. It was concluded that a gradual decrease of fill ratio, observed in the free flowing materials, was due to higher particle inertia that prevents them being drawn inside the die as a single compacted column.

These results are interesting in relation to paddle feeding, used in most tabletting machines. During the rotation of paddles inside the feed shoe, powder particles are subjected to significant accelerations during their transit over the dies. According to the results of this study, cohesive powders might require higher paddle velocities in order to densify the powder mass and ensure a uniform plug flow. Free-flowing materials, on the other hand, might perform better at lower paddle velocities. Further investigations need to be carried out in order to verify the correlation between the filling performance in linear systems and actual tablet presses.

Acknowledgements

IFPRI, International Fine Particle Research Institute, USA.

References

[1] C.-Y. Wu, L. Dihoru, A.C.F. Cocks, The flow of powder into simple and stepped dies, Powder Technol. 134 (1) (2003) 24–39.
[2] C.-Y. Wu, A.C.F. Cocks, Flow behaviour of powders during die filling, Powder Metall. 47 (2) (2004) 127–136.
[3] L. Schneider, I.C. Sinka, A.C.F. Cocks, Characterisation of the flow behaviour of pharmaceutical powders using a model die–shoe filling system, Powder Technol. 173 (1) (2007) 59–71.
[4] L. Schneider, I.C. Sinka, A.C.F. Cocks, Characterisation of the flow behaviour of pharmaceutical powders using a model die–shoe filling system, Powder Technol. 173 (1) (2007) 59–71.
[5] C.-Y. Wu, B. Armstrong, N. Vlachos, Characterization of Powder Flowability for Die Filling, 2012.
[6] C.-Y. Wu, A.C.F. Cocks, O.T. Gillia, D.A. Thompson, Experimental and numerical investigations of powder transfer, Powder Technol. 138 (2–3) (2003) 216–228.
[7] I.C. Sinka, L. Schneider, A.C.F. Cocks, Measurement of the flow properties of powders with special reference to die fill, Int. J. Pharm. 280 (1) (2004) 27–38.
[8] I.C. Sinka, F. Motazedian, A.C.F. Cocks, K.G. Pitt, The effect of processing parameters on pharmaceutical tablet properties, Powder Technol. 189 (2) (2009) 276–284.
[9] L. Schneider, A.C.F. Cocks, A. Apostolopoulos, Ceramic comparison of filling behaviour of metallic, ceramic, hardmetal and magnetic powders, Powder Metall. 48 (1) (2005) 77–84.
[10] H.P. Goh, P.W.S. Heng, C.V. Liew, Understanding effects of process parameters and forced feeding on die filling, Eur. J. Pharm. Sci. 122 (2018) 105–115.
[11] S. Jackson, I.C. Sinka, A.C.F. Cocks, The effect of suction during die fill on a rotary tablet press, Eur. J. Pharm. Biopharm. 65 (2) (2007) 253–256.
[12] C.-Y. Wu, Yu Guo, Numerical modelling of suction filling using DEM/CFD, Chem. Eng. Sci. 73 (2012) 231–238.
[13] Y. Guo, C.-Y. Wu, K.D. Kafui, C. Thornton, 3D DEM/CFD analysis of size-induced segregation during die filling, Powder Technol. 206 (1) (2011) 177–188.
[14] GranuFlow Granular Material Flowmeter, (N.d.) Retrieved from https://granutools.com/products/granuflow/ 2019.
[15] X. Fu, D. Huck, L. Makein, B. Armstrong, U. Willen, T. Freeman, Effect of particle shape and size on flow properties of lactose powders, Particuology 10 (2) (2012) 203–208.
[16] M. Hertel, E. Schwarz, M. Kobler, S. Hauptstein, H. Steckel, R. Scherließ, Powder flow analysis: a simple method to indicate the ideal amount of lactose fines in dry powder inhaler formulations, Int. J. Pharm. 535 (1–2) (2018) 59–67.
[17] S. Mohammed, E. Abdullah, D. Geldart, A. Raman, Measuring powder flowability with a modified Warren spring cohesion tester, Particuology 9 (2) (2011) 148–154.
[18] E. Gnagne, H. Petit, J. Gaiani, C. Scher, G. Amani, Characterisation of flow properties of foutou and foufou flours, staple foods in West Africa, using the FT4 powder rheometer, J. Food Meas. Charact. 11 (3) (2017) 1128–1136.
[19] M. Leturia, M. Benali, S. Lagarde, I. Ronga, K. Saleh, Characterization of flow properties of cohesive powders: a comparative study of traditional and new testing methods, Powder Technol. 253 (2014) 406–423.
[20] K.S. Babu, K. Siliveru, J.K. Amamcharla, P.V. Vadlani, R.P. Kingsly Ambrose, Influence of protein content and storage temperature on the particle morphology and flowability characteristics of milk protein concentrate powders, J. Dairy Sci. 101 (8) (2018) 7013–7026.